

Herramientas ontológicas 1

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Construcción de ontologías en Ontology Web Language (OWL) con Protege

Construcción de ontologías en OWL

- **Classes**
- **Properties**
 - **Object properties**
 - **Datatype properties**
 - **Annotations**
- **Axioms (Restrictions, asserted conditions)**
- **Individuals**

Construcción de ontologías en OWL

- **Terminological knowledge (T-BOX):**
 - **Classes**
 - **Properties**
 - **Object properties**
 - **Datatype properties**
 - **Annotations**
 - **Individuals**
 - **Axioms (Restrictions, asserted conditions)**
- **Knowledge about individuals (A-BOX):**
 - **Individuals**

TBOX

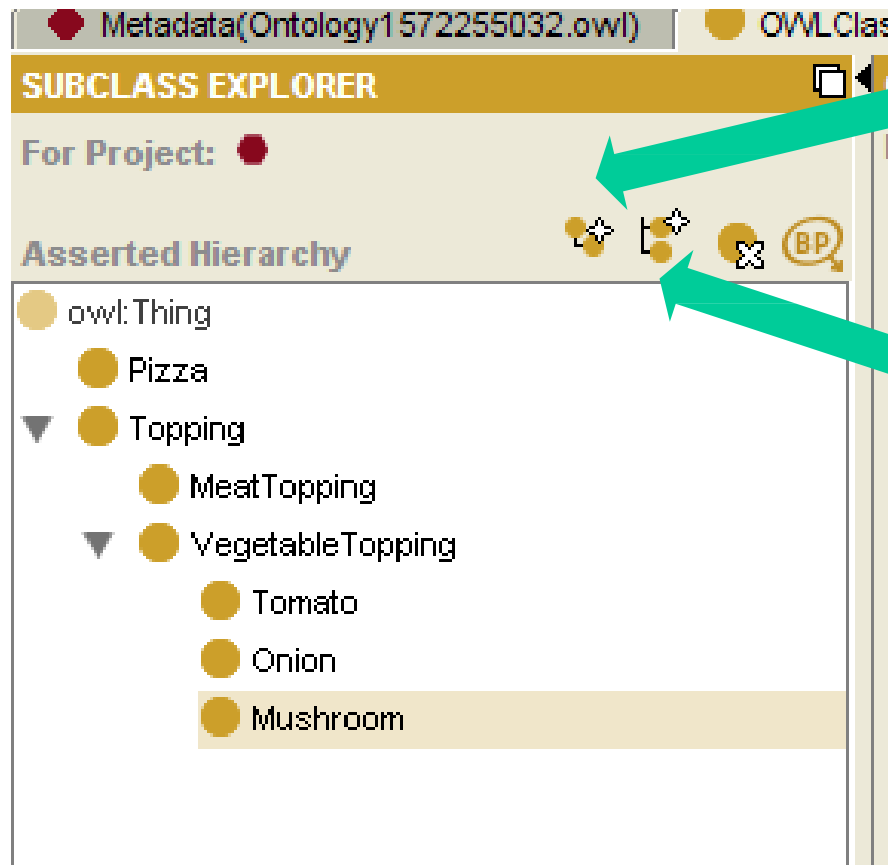
Editores ontológicos (en OWL): Protege

- **SWOOP**
- **Protégé 3. Stanford**
- **Protége 4**
- ...

Axiomas en OWL

Axiom	DL Syntax	Example
subClassOf	$C_1 \sqsubseteq C_2$	Human $\sqsubseteq$ Animal $\sqcap$ Biped
equivalentClass	$C_1 \equiv C_2$	Man $\equiv$ Human $\sqcap$ Male
disjointWith	$C_1 \sqsubseteq \neg C_2$	Male $\sqsubseteq \neg$ Female
sameIndividualAs	$\{x_1\} \equiv \{x_2\}$	{President_Bush} $\equiv$ {G_W_Bush}
differentFrom	$\{x_1\} \sqsubseteq \neg\{x_2\}$	{john} $\sqsubseteq \neg\{peter\}$
subPropertyOf	$P_1 \sqsubseteq P_2$	hasDaughter $\sqsubseteq$ hasChild
equivalentProperty	$P_1 \equiv P_2$	cost $\equiv$ price
inverseOf	$P_1 \equiv P_2^-$	hasChild $\equiv$ hasParent ⁻
transitiveProperty	$P^+ \sqsubseteq P$	ancestor ⁺ $\sqsubseteq$ ancestor
functionalProperty	$\top \sqsubseteq \leq 1P$	$\top \sqsubseteq \leq 1$ hasMother
inverseFunctionalProperty	$\top \sqsubseteq \leq 1P^-$	$\top \sqsubseteq \leq 1$ hasSSN ⁻

Creación de classes



• Create new child

• Create new sibling

Creación de properties: convenciones

- **Notación**
 - Letra minúscula para comenzar
- **Las properties se crean usando 2 patrones posibles estándar :**
 - has... (hasTopping, hasChild)
 - is...Of (isToppingOf) u otros sufijos (In o bien To)
- **Ventajas:**
 - Resulta más fácil encontrar las properties (a través de expresiones regulares)
- **Las properties inversas normalmente siguen el siguiente patrón: hasPart, isPartOf**

Adición de datatype properties (atributos)

PROPERTY BROWSER

For Project:

Object Datatype Annotation All

Datatype Properties

- price

PROPERTY EDITOR for price (instance of owl:DatatypeProperty)

For Property: <http://www.owl-ontologies.com/Ontology1572255032.owl#price>

Property	Value
rdfs:comment	

Select the Domain and Range.

Domain

- Pizza

Range

int optional

Allowed values

Adición de object properties (relaciones)

The screenshot displays the 'PROPERTY EDITOR for hasTopping (instance of owl:ObjectProperty)' window. The 'For Property' field is set to `http://www.owl-ontologies.com/Ontology1572255032.owl#hasTopping`. The 'Object properties' list on the left shows `isToppingOf ↔ hasTopping` and `hasTopping ↔ isToppingOf`. The 'Domain' is set to 'Pizza' and the 'Range' is set to 'Topping'. The 'Inverse' checkbox is checked, and the 'Transitive' checkbox is unchecked. The 'Functional' and 'InverseFunctional' checkboxes are also unchecked. The 'Lang' field is empty.

For Project:

Object Datatype Annotation All

Object properties

- isToppingOf ↔ hasTopping
- hasTopping ↔ isToppingOf

Property Value Lang

rdfs:comment		
--------------	--	--

Select the Domain and Range.

Domain Range

- Pizza
- Topping

☐ Functional
☐ InverseFunctional
☒ Inverse
☐ Symmetric
☐ Transitive

Inverse
<deleted frame>

• Select inverse relationship if it is needed

Properties de las relaciones en OWL 1.1

- **FunctionalProperty**
 - $(X \text{ Relation } Y) \text{ and } (X \text{ Relation } Z) \rightarrow (Y = Z)$
- **InverseFunctionalProperty**
 - $(X \text{ Relation } Y) \text{ and } (Z \text{ Relation } Y) \rightarrow (X = Z)$
- **Symmetry**
 - $(X \text{ Relation } Y) \leftrightarrow (Y \text{ Relation } X)$
- **Transitive**
 - $(X \text{ Relation } Y) \text{ and } (Y \text{ Relation } Z) \rightarrow (X \text{ Relation } Z)$
- **Inverse**

Properties y herencia

- Las Properties se heredan de las supraclases

The screenshot displays an ontology editor interface with two main panels. The left panel, titled 'SUBCLASS EXPLORER', shows an 'Asserted Hierarchy' for a project. The hierarchy is as follows:

- owl:Thing
 - Pizza
 - Topping
 - MeatTopping
 - Pork (highlighted)
 - VegetableTopping
 - Tomato
 - Onion
 - Mushroom

The right panel, titled 'CLASS EDITOR for Pork (instance of owl:Class)', shows the 'For Class' as `http://www.owl-ontologies.com/Ontology1572255032.owl#Pork`. It contains a table with two columns: 'Property' and 'Value'. The table lists the following properties:

Property	Value
<code>rdfs:comment</code>	

 Below the table, there are two additional properties listed:

- `calories` (multiple int)
- `isToppingOf` (multiple Pizza)

Adición de individuos (instancias)

The screenshot displays an OWL editor interface with three main panels:

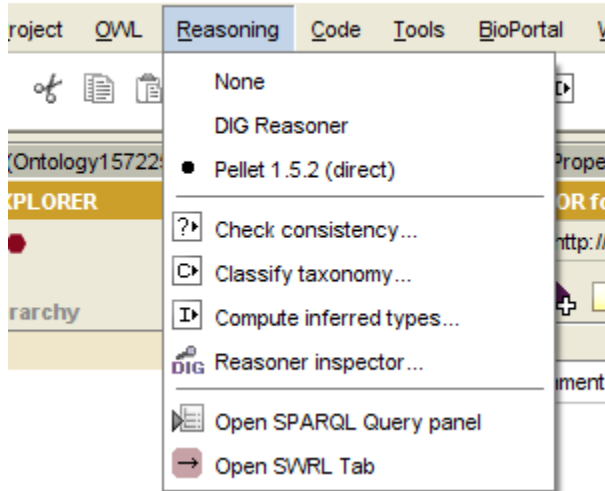
- CLASS BROWSER**: Shows a class hierarchy for the project. The hierarchy is as follows:
 - owl:Thing
 - Pizza
 - Topping
 - MeatTopping
 - VegetableTopping
 - Tomato (1)
 - Onion
 - Mushroom

- INSTANCE BROWSER**: Shows the 'Asserted' tab for the class 'Tomato'. It lists one instance: 'TomatoCherryCarrefour'.
- INDIVIDUAL EDITOR for Tomato**: Shows the 'For Individual' field with the value 'jy1572255032.ov'. Below it is a table with one row:

Property	
rdfs:comment	

Razonamiento en OWL

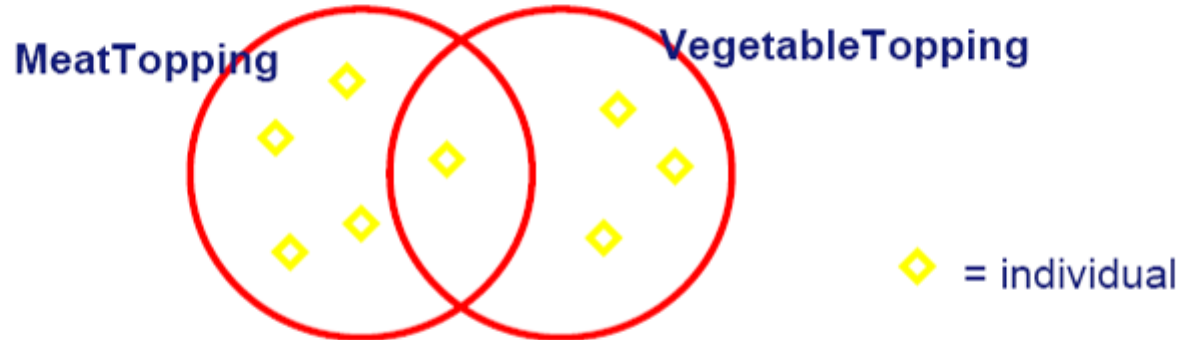
Protégé 3.5



- **Chequeo de consistencia**
 - Clases disjuntas
 - Restricciones
- **Taxonomy classification**
- **Individuals classification**
(computación de tipos inferidos –p.ej. por transitividad)

Disjointness

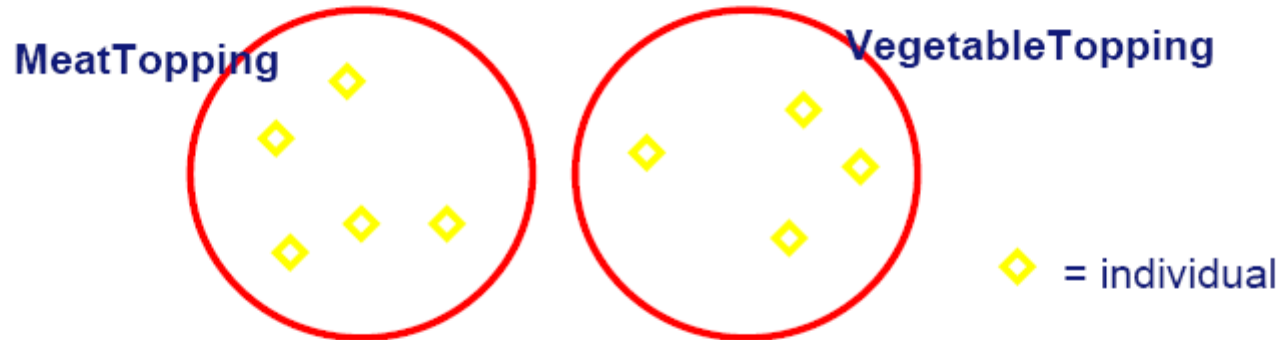
- OWL assumes that classes overlap



- ▶ This means an individual could be both a **MeatTopping** and a **VegetableTopping** at the same time
- ▶ We want to state this is not the case

Disjointness

- If we state that classes are disjoint

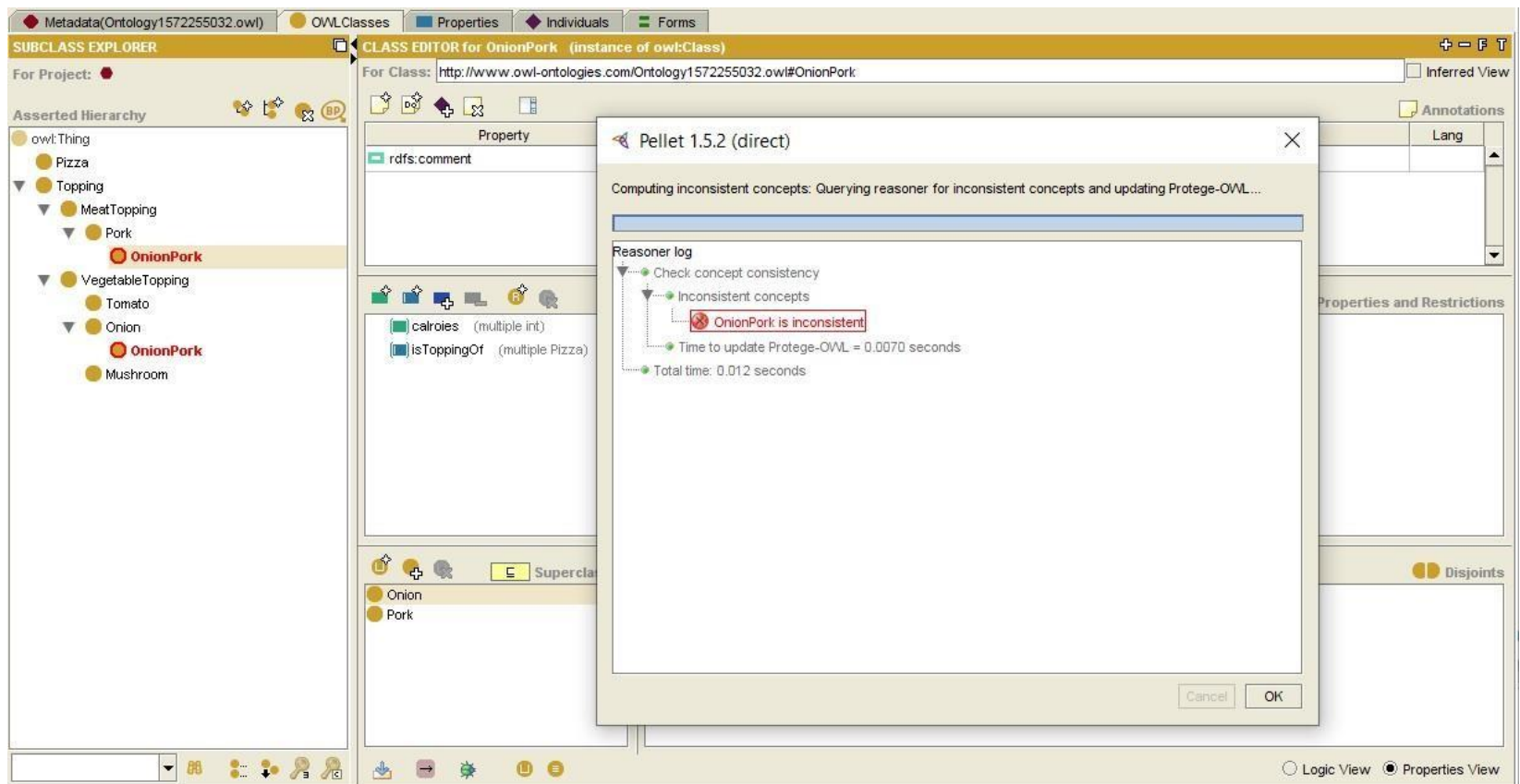


- ▶ This means an individual cannot be both a **MeatTopping** and a **VegetableTopping** at the same time
- ▶ We must do this explicitly in the interface

ClassesTab: Disjoints Widget



Chequeo de inconsistencias



Creación de restricciones

The screenshot displays the Protégé 3.5 software interface. The main window is titled "<new> Protégé 3.5". The menu bar includes File, Edit, Project, OWL, Reasoning, Code, Tools, BioPortal, Window, Collaboration, and Help. The toolbar contains various icons for file operations and editing. The interface is divided into several panes:

- Subclass Explorer:** Located on the left, it shows an asserted hierarchy starting from `owl:Thing`, followed by `Pizza`, `NamedPizza`, and `AmericanPizza`. Other classes like `Topping`, `MeatTopping`, `Pork`, `VegetableTopping`, `Tomato`, `Onion`, and `Mushroom` are also listed.
- Class Editor:** The central pane is titled "CLASS EDITOR for AmericanPizza (instance of owl:Class)". It shows the URI `http://www.owl-ontologies.com/Ontology1572255032.owl#AmericanPizza` and an "Inferred View" checkbox. Below this is a table with columns "Property", "Value", and "Lang". The first row shows `rdfs:comment` in the "Property" column.
- Properties and Restrictions:** The bottom pane shows a list of properties: `hasTopping` (multiple Topping) and `price` (multiple). A context menu is open over the `hasTopping` property, offering options to create various restrictions:
 - Create allValuesFrom restriction
 - Create someValuesFrom restriction
 - Create hasValue restriction
 - Create cardinality restriction
 - Create minCardinality restriction
 - Create maxCardinality restriction
 - Delete property
 - Change property metaclass...
 - Convert to owl:DatatypeProperty
 - Refactor
 - Search and View
- Disjoints:** A section on the right side of the bottom pane, currently empty.

The status bar at the bottom indicates "Logic View" and "Properties View".

Restricciones posibles

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
someValuesFrom	$\exists$	some	hasChild some Man
allValuesFrom	$\forall$	only	hasSibling only Woman
hasValue	$\ni$	value	hasCountryOfOrigin value England
minCardinality	$\geq$	min	hasChild min 3
cardinality	$=$	exactly	hasChild exactly 3
maxCardinality	$\leq$	max	hasChild max 3

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
intersectionOf	$\sqcap$	and	Doctor and Female
unionOf	$\sqcup$	or	Man or Woman
complementOf	$\neg$	not	not Child

Ejemplo de restricciones (I)

- SomeValuesFrom

The screenshot shows an OWL editor interface with two main panes. The left pane, titled 'SUBCLASS EXPLORER', displays a hierarchy of classes for the project 'pizza'. The right pane, titled 'CLASS EDITOR for AmericanPizza', shows the details for the class 'AmericanPizza'. A green circle highlights the 'hasTopping' property restriction in the class editor.

SUBCLASS EXPLORER

For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza
 - PizzaBase
 - Topping

CLASS EDITOR for AmericanPizza (instance of owl:Class)

For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#AmericanPizza>

Property
☒ rdfs:comment

hasTopping (multiple Topping) (someValuesFrom Parmesan, s)

- Parmesan
- RoquefortCheese
- Mushrooms
- Chicken
- Beef
- Bacon

hasPizzaBase (multiple PizzaBase)

price (multiple float)

Superclasses

- NamedPizza

Ejemplo de restricciones (II)

- **AllValuesFrom**

The screenshot displays a web ontology editor interface. On the left, the 'SUBCLASS EXPLORER' shows a hierarchy of classes under 'pizza', with 'OnlyMushroomPizza' selected. The main 'CLASS EDITOR for OnlyMushroomPizza' shows the class's properties. The 'hasTopping' property is highlighted with a green circle, showing its restriction: '(allValuesFrom Mushrooms)'. The 'price' property is also visible, restricted to '(multiple float)'. The 'Superclasses' section at the bottom lists 'NamedPizza'.

Metadata(Ontology1509470563.owl) | OWLClasses | Properties | Individuals | Forms

SUBCLASS EXPLORER

For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza**
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza
 - PizzaBase
 - Topping

CLASS EDITOR for OnlyMushroomPizza (instance of owl:Class)

For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#OnlyMushroomPizza>

Property	Value
rdfs:comment	

hasPizzaBase (multiple PizzaBase) (someValuesFrom CheeseBorderBase)

- CheeseBorderBase

hasTopping (allValuesFrom Mushrooms)

- Mushrooms

price (multiple float)

Superclasses

- NamedPizza

Ejemplo de restricciones (III)

- **hasValue**

Metadata(Ontology1509470563.owl) | OWLClasses | Properties | Individuals | Forms

SUBCLASS EXPLORER
For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour**
 - SeaFoodPizza
 - VegetarianPizza
 - PizzaBase
 - Topping

CLASS EDITOR for PizzaWithTomatoCherryCarrefour (instance of owl:Class)
For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#PizzaWithTomatoCherryCarrefour>

Property	Value
rdfs:comment	

hasTopping (multiple Topping) (hasValue TomatoCherryCarrefour)
TomatoCherryCarrefour
hasPizzaBase (multiple PizzaBase)
price (multiple float)

Defining Classes

Pizza

Ejemplo de restricciones (IV)

- Cardinality restrictions

The screenshot displays the Protege OWL editor interface. On the left, the 'SUBCLASS EXPLORER' shows a hierarchy for the 'pizza' project, with 'ExpensivePizza' selected under the 'Pizza' class. The main area is the 'CLASS EDITOR for ExpensivePizza', which shows the 'hasTopping' property with a cardinality restriction of '≥ 5' (multiple Topping, minCardinality 5). This restriction is highlighted with a green circle. The 'CLASS EDITOR' also shows other properties like 'hasPizzaBase' and 'price'. The bottom of the editor shows the 'Defining Classes' tab with 'Pizza' selected.

Metadata(Ontology1509470563.owl) | OWLClasses | Properties | Individuals | Forms

SUBCLASS EXPLORER
For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza
- PizzaBase
- Topping

CLASS EDITOR for ExpensivePizza (instance of owl:Class)
For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl>

Property
rdfs:comment

hasTopping (multiple Topping) (minCardinality 5)

- hasPizzaBase (multiple PizzaBase)
- price (multiple float)

Defining Classes

- Pizza

Properties vs Logic View

The screenshot displays the Protégé OWL editor interface. The top tabs include Metadata, OWLClasses, Properties, Individuals, and Forms. The SUBCLASS EXPLORER on the left shows a hierarchy starting from owl:Thing, with Pizza as a subclass. Under Pizza, there are several subclasses: CheapPizza, CheesyPizza, ExpensivePizza, MuslimForbiddenPizza, MuslimPermittedPizza, NamedPizza, and others. The CLASS EDITOR for AmericanPizza (instance of owl:Class) is the active window. It shows the URI for the class and an Inferred View checkbox. Below this is a table with columns for Property, Value, and Lang. The table contains one row: rdfs:comment. Below the table is a section for Asserted Conditions, which is currently empty. At the bottom right, there are two radio buttons: Logic View (selected) and Properties View. The Logic View is circled in green.

Metadata(Ontology1509470563.owl) | OWLClasses | Properties | Individuals | Forms

SUBCLASS EXPLORER

For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza
 - PizzaBase
 - Topping

CLASS EDITOR for AmericanPizza (instance of owl:Class)

For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#AmericanPizza> ☐ Inferred View

Property	Value	Lang
rdfs:comment		

Asserted Conditions

NECESSARY & SUFFICIENT

NECESSARY

- NamedPizza
 - hasTopping some Parmesan
 - hasTopping some RoquefortCheese
 - hasTopping some Mushrooms
 - hasTopping some Chicken
 - hasTopping some Beef
 - hasTopping some Bacon

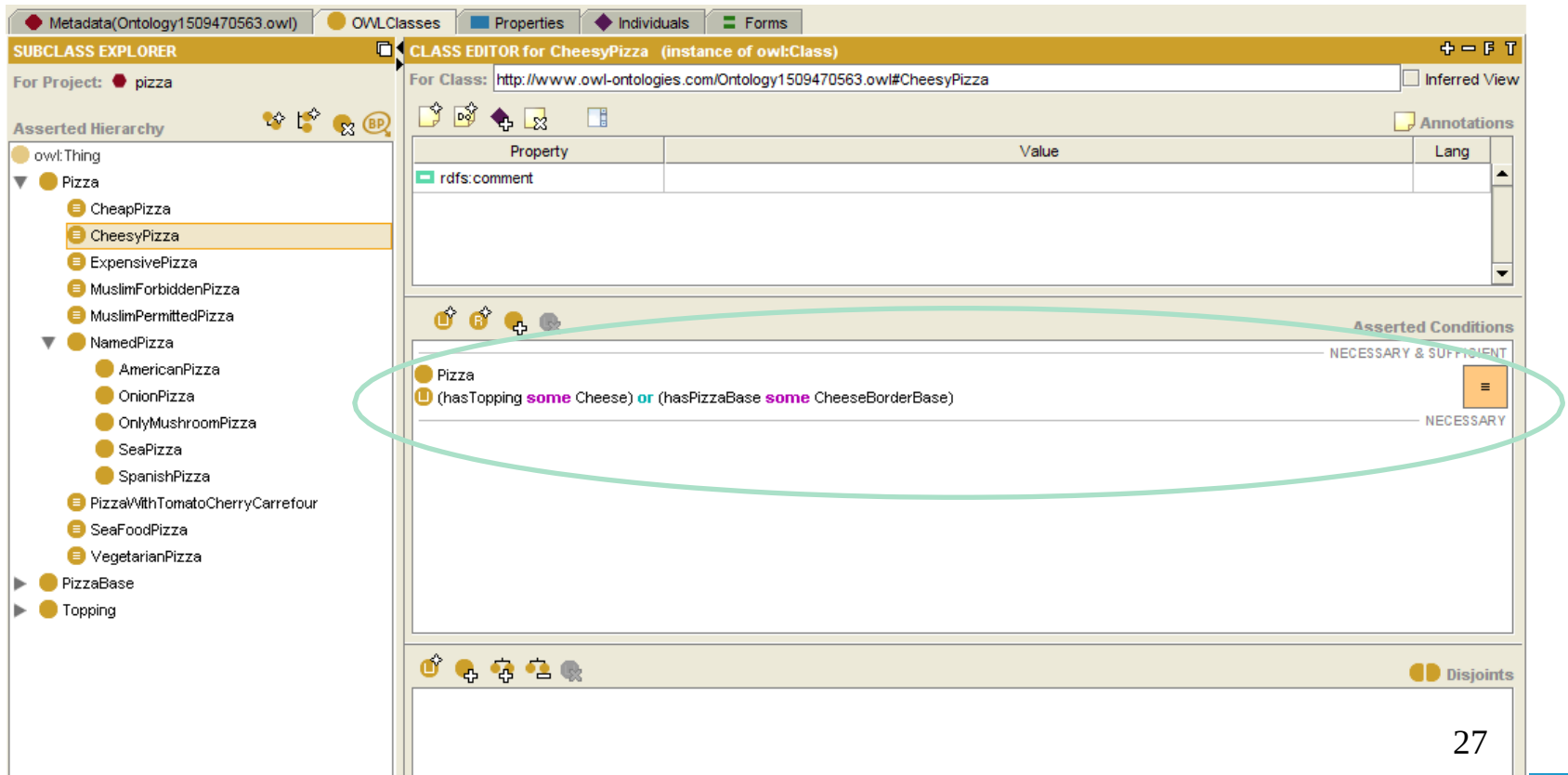
Disjoints

☒ Logic View ☐ Properties View

Condición necesaria y suficiente

- An Equivalent class has «Necessary and sufficient conditions»

 CheesyPizza



Metadata(Ontology1509470563.owl) | OWLClasses | Properties | Individuals | Forms

SUBCLASS EXPLORER
For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza**
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza
- PizzaBase
- Topping

CLASS EDITOR for CheesyPizza (instance of owl:Class)
For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#CheesyPizza> ☐ Inferred View

Property	Value	Lang
 rdfs:comment		

Asserted Conditions

-  Pizza
-  (hasTopping **some** Cheese) or (hasPizzaBase **some** CheeseBorderBase)

NECESSARY & SUFFICIENT

NECESSARY

Disjoints

Clasificación taxonómica

Metadata(Ontology1509470563.owl) OWLClasses Properties Individuals Forms

SUBCLASS EXPLORER

For Project: pizza

Asserted Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - VegetarianPizza

SUBCLASS EXPLORER

For Project: pizza

Inferred Hierarchy

- owl:Thing
 - Pizza
 - CheapPizza
 - CheesyPizza
 - AmericanPizza
 - OnionPizza
 - OnlyMushroomPizza
 - SeaPizza
 - ExpensivePizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - NamedPizza
 - PizzaWithTomatoCherryCarrefour
 - SeaFoodPizza
 - PizzaBase
 - Topping

CLASS EDITOR for CheesyPizza (instance of owl:Class)

For Class: <http://www.owl-ontologies.com/Ontology1509470563.owl#Cheesy>

Property	Value
rdfs:comment	

☐ Pizza
☒ (hasTopping **some** Cheese) **or** (hasPizzaBase **some** CheeseBorderBase)

Class	Changed direct superclasses
AmericanPizza	Added CheesyPizza, ExpensivePizza, MuslimForbiddenPizza
OnionPizza	Added CheesyPizza, VegetarianPizza, CheapPizza
OnlyMushroomPizza	Added CheesyPizza, VegetarianPizza
SeaPizza	Added CheesyPizza, SeaFoodPizza
SpanishPizza	Added MuslimForbiddenPizza
VegetarianPizza	Moved from Pizza to MuslimPermittedPizza

Manchester OWL Syntax

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
someValuesFrom	$\exists$	some	hasChild some Man
allValuesFrom	$\forall$	only	hasSibling only Woman
hasValue	$\ni$	value	hasCountryOfOrigin value England
minCardinality	$\geq$	min	hasChild min 3
cardinality	$=$	exactly	hasChild exactly 3
maxCardinality	$\leq$	max	hasChild max 3

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
intersectionOf	$\sqcap$	and	Doctor and Female
unionOf	$\sqcup$	or	Man or Woman
complementOf	$\neg$	not	not Child

Mundo abierto vs cerrado

- **Close world**
 - Like databases
 - «Any statement that is not known is **false**»
- **Open world**
 - Ontologies
 - «The knowledge cannot be completed»
 - «Any statement that is not known is **not known**»

Asserted Polyhierarchies

We believe asserting polyhierarchies is bad

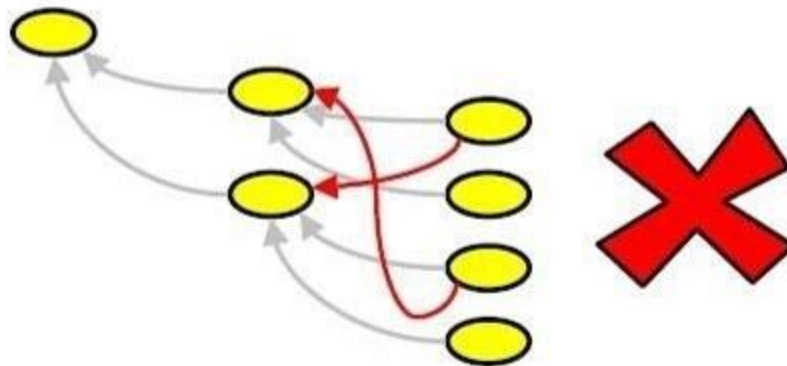
We lose some encapsulation of knowledge

Why is this class a subclass of that one?

Difficult to maintain

Adding new classes becomes difficult because all subclasses may need to be updated

Extracting from a graph is harder than from a tree



let the reasoner do it!

Bibliografía

- **Ontology Development 101: A Guide to Creating Your First Ontology**
 - http://protege.stanford.edu/publications/ontology_development/ontology101.pdf

Herramientas ontológicas 2

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Protege 5.5 y OVL 2

Manchester OWL Syntax

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
someValuesFrom	$\exists$	some	hasChild some Man
allValuesFrom	$\forall$	only	hasSibling only Woman
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minCardinality	$\geq$	min	hasChild min 3
cardinality	$=$	exactly	hasChild exactly 3
maxCardinality	$\leq$	max	hasChild max 3

OWL	DL Symbol	Manchester OWL Syntax Keyword	Example
intersectionOf	$\sqcap$	and	Doctor and Female
unionOf	$\sqcup$	or	Man or Woman
complementOf	$\neg$	not	not Child

Protege 5.5

protege5.5.2 (http://www.owl-ontologies.com/pizza1572370159.owl) : [C:\Users\RafaelValenciaGarcia\Downloads\pizza.owl]

File Edit View Reasoner Tools Refactor Window Help

< > protege5.5.2 (http://www.owl-ontologies.com/pizza1572370159.owl) Search...

Pizza > CheesyPizza

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: CheesyPizza Annotations: CheesyPizza

Class Annotations Class Usage

Annotations +

Description: CheesyPizza

Equivalent To +

Pizza and ((hasBase some CheeseBorderBase) or (hasTopping some Cheese))

SubClass Of +

Pizza

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Pizza_39

Target for Key +

Disjoint With +

Disjoint Union Of +

owl:Thing

- Pizza
 - CheesyPizza
 - HeavyPizza
 - LightPizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - MuslimPermittedPizza2
 - NamedPizza
 - AmericanPizza
 - FourCheesePizza
 - OnionPizza
 - OnlyMushroomsPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - VeganPizza
- PizzaBase
 - CheeseBorderBase
 - DeepPanBase
 - SlimPizzaBase
- Topping
 - Cheese
 - FishAndSeafood
 - MeatTopping
 - VegetableTopping

OWL 2

OWL 2 (1.1) añade, respecto a OWL1:

- **qualified cardinality restrictions**, como `ObjectMinCardinality(2 friendOf hacker)`;
- **local reflexivity restrictions**, on noncomplex properties only, such as `ObjectExistsSelf(likes)` for narcissists;
- **reflexive, irreflexive, symmetric, and asymmetric properties**, for noncomplex properties only, such as `ReflexiveObjectProperty(knows)` and `IrreflexiveObjectProperty(husbandOf)`;
- **disjoint properties**, for non-complex properties only, such as `DisjointObjectProperties(childOf spouseOf)`; and
- **property chain inclusion axioms**, such as `SubObjectPropertyOf(SubObjectPropertyChain(owns part) owns)`, provided that there are no cyclic inclusions.

Razonamiento

The screenshot displays the Protégé OWL editor interface. The title bar shows the file path: `pizza1572370159 (http://www.owl-ontologies.com/pizza1572370159.owl) : [C:\Users\RafaelValenciaGarcia\Downloads\pizza.owl]`. The menu bar includes File, Edit, View, Reasoner, Tools, Refactor, Window, and Help. The Reasoner menu is open, showing options: Start reasoner, Synchronize reasoner, Stop reasoner, Explain inconsistent ontology, and Configure... The Configure... option is selected, and a sub-menu is visible with the following items: Hermit 1.4.3.456 (highlighted with a green circle) and None. The main workspace shows the class hierarchy for `CheesyPizza`. The hierarchy is as follows: `owl:Thing` is the root, with `Pizza` as a child. `Pizza` has several children: `AmericanPizza`, `FourCheesePizza`, `OnionPizza`, `SeaPizza`, `SpanishPizza`, `HeavyPizza`, `LightPizza`, `MuslimForbiddenPizza`, `MuslimPermittedPizza`, `MuslimPermittedPizza2`, `NamedPizza`, `PizzaWithTomatoCherryCarrefour`, `PizzaBase`, and `Topping`. The `CheesyPizza` class is highlighted in blue. The right pane shows the description of `CheesyPizza`, which is equivalent to `Pizza` and `((hasBase some CheeseBorderBase) or (hasTopping some Cheese))`. The `SubClass Of` section shows `Pizza` as a subclass of `CheesyPizza`. The `Instances` section shows `Pizza_39` as an instance of `CheesyPizza`. The bottom status bar indicates "Reasoner active" and "Show Inferences" is checked.

Reasoner active ☒ Show Inferences

Creación de clases

The screenshot displays the Protégé ontology editor interface. The main window shows the 'Country' class being edited. The left pane, titled 'Class hierarchy: Country', shows a tree structure where 'Country' is a subclass of 'Location', which is a subclass of 'owl:Thing'. Other subclasses of 'Location' include 'City', 'Suburb', and 'Person'. The right pane, titled 'Description: Country', shows the 'Equivalent To' section with a plus sign, and the 'SubClass Of' section with a plus sign and a list containing 'Location'. The 'General class axioms' section also has a plus sign. The bottom status bar indicates 'To use the reasoner click Reasoner > Start reasoner' and 'Show Inferences'.

untitled-ontology-39 (http://www.semanticweb.org/rafaelvalenciagarcia/ontologies/2019/10/untitled-ontology-39) : [http://www.semanticweb.org/rafaelvale...

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< > untitled-ontology-39 (http://www.semanticweb.org/rafaelvalenciagarcia/ontologies/2019/10/untitled-ontology-39) Search...

/Location /Country

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: Country

owl:Thing

- Location
 - Country
 - City
 - Suburb
- Person

Class Annotations Class Usage

Annotations: Country

Annotations +

Description: Country

Equivalent To +

SubClass Of +

- Location

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

Disjoint Union Of +

To use the reasoner click Reasoner > Start reasoner Show Inferences

Adición de datatype properties

The screenshot displays the Protégé ontology editor interface. The main window shows the configuration for a datatype property named 'age'.

Left Panel (Data property hierarchy: age):

- Annotation properties
- Datatypes
- Individuals
- Classes
- Object properties
- Data properties

Right Panel (Annotations: age):

- Annotations
- Data Property Usage

Bottom Panel (Description: age):

- ☐ Functional
- Equivalent To: +
- SubProperty Of: +
- Domains (intersection): +
 - Person
- Ranges: +
 - xsd:int
- Disjoint With: +

Two green ovals highlight the 'Domains (intersection)' and 'Ranges' sections, indicating the addition of the 'Person' domain and the 'xsd:int' range to the 'age' property.

Adición de object properties

The screenshot shows the Protégé ontology editor interface. The main window displays the 'livesIn' object property configuration. The left sidebar shows the 'Object property hierarchy: livesIn' with 'owl:topObjectProperty' and 'livesIn' listed. The right sidebar shows the 'Description: livesIn' with various property constraints. A green oval highlights the 'Person' and 'Location' classes under the 'Domains (intersection)' and 'Ranges (intersection)' sections. The bottom right corner has a green oval with the number '4' and a small '1' below it.

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livesIn

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Object property hierarchy: livesIn

Annotations: livesIn

Annotations +

Characteristics: livesIn

Description: livesIn

Equivalent To +

SubProperty Of +

Inverse Of +

Domains (intersection) +

Person

Ranges (intersection) +

Location

Disjoint With +

SuperProperty Of (Chain) +

4

1

Object properties characteristics

The screenshot displays the Protégé ontology editor interface. The main window shows the 'Object property hierarchy' for the 'livesIn' property, which is currently set to 'Asserted'. The hierarchy shows 'owl:topObjectProperty' as the parent and 'livesIn' as the child.

Below the hierarchy, the 'Object property characteristics' panel is visible, listing several characteristics with checkboxes:

- ☐ Functional
- ☐ Inverse functional
- ☐ Transitive
- ☐ Symmetric
- ☐ Asymmetric
- ☐ Reflexive
- ☐ Irreflexive

The 'Description: livesIn' panel on the right shows the property's domain and range. The domain is 'Person' and the range is 'Location'. The 'Equivalent To' and 'SubProperty Of' sections are empty.

The 'SuperProperty Of (Chain)' section at the bottom is highlighted in blue.

OWL 2

- **Reflexive (*R*).**
 - $(X \text{ Relation } X)$
- **Irreflexive (*I*).**
 - $\text{not}(X \text{ Relation } X).$
- **Symmetric (*S*).**
 - $(X \text{ Relation } Y) \leftrightarrow (Y \text{ Relation } X)$
- **Asimmetric (*A*).**
 - $(X \text{ Relation } Y) \rightarrow \text{not}(Y \text{ Relation } X).$
- **Transitive (*T*).**
 - $(X \text{ Relation } Y) \text{ and } (Y \text{ Relation } Z) \rightarrow (X \text{ Relation } Z).$
- **Functional (*F*).**
 - $(X \text{ Relation } Y) \text{ and } (X \text{ Relation } Z) \rightarrow (Y = Z)$
- **Inverse Functional (*Inv .F*).**
 - $(X \text{ Relation } Y) \text{ and } (Z \text{ Relation } Y) \rightarrow (X = Z)$

Adición de individuals

The screenshot displays the Protégé ontology editor interface for an ontology named 'untitled-ontology-39'. The main window is divided into several panes:

- Class hierarchy:** Shows a tree structure starting from 'owl:Thing', with 'Location' as a subclass. 'Location' has three subclasses: 'City', 'Country', and 'Suburb'. 'Person' is a subclass of 'Location' and is currently selected.
- Annotations: Rafa:** A pane for adding annotations to the individual 'Rafa'. It contains a search bar and a list of annotations.
- Description: Rafa:** A pane showing the description of the individual 'Rafa'. It lists the types 'Person' and 'Location'.
- Property assertions: Rafa:** A pane for adding property assertions to the individual 'Rafa'. It contains sections for 'Object property assertions', 'Data property assertions', 'Negative object property assertions', and 'Negative data property assertions'. The 'Object property assertions' section shows the assertion 'livesIn Murcia'. The 'Data property assertions' section shows the assertion 'age "41"^^xsd:int'.
- Direct instances: Rafa:** A pane for adding direct instances to the individual 'Rafa'. It shows the instance 'Rafa'.

The 'Person' class is highlighted in the class hierarchy, and the 'Rafa' individual is listed in the 'Direct instances' pane. The 'livesIn Murcia' and 'age "41"^^xsd:int' assertions are visible in the 'Property assertions' pane.

Condiciones necesarias

Visual representation of an OWL ontology editor showing the class hierarchy and the 'SubClassOf' relationship for the 'AmericanPizza' class.

The interface displays the following structure:

- Active ontology: pizza1572370159 (http://www.owl-ontologies.com/pizza1572370159.owl)
- Class hierarchy: AmericanPizza
 - owl:Thing
 - Pizza
 - CheesyPizza
 - HeavyPizza
 - LightPizza
 - MuslimForbiddenPizza
 - MuslimPermittedPizza
 - MuslimPermittedPizza2
 - NamedPizza
 - AmericanPizza**
 - FourCheesePizza
 - OnionPizza
 - OnlyMushroomsPizza
 - SeaPizza
 - SpanishPizza
 - PizzaWithTomatoCherryCarrefour
 - VeganPizza
 - PizzaBase
 - CheeseBorderBase
 - DeepPanBase
 - SlimPizzaBase
 - Topping
 - Cheese
 - FishAnd Seafood
 - MeatTopping
 - VegetableTopping

The 'SubClassOf' relationship for 'AmericanPizza' is shown in the right pane, listing the following classes as subclasses:

- hasTopping some Bacon
- hasTopping some Beef
- hasTopping some MozzarellaCheese
- hasTopping some ParmesanCheese
- hasTopping some Tomato
- hasTopping some Tuna
- NamedPizza
- CheesyPizza
- HeavyPizza
- MuslimForbiddenPizza

The 'SubClassOf' relationship is highlighted with a green circle.

Condiciones necesarias y suficientes

The screenshot shows the Protégé OWL editor interface. On the left, the 'Class hierarchy: CheesyPizza' panel displays a tree structure starting from 'owl:Thing' and branching into 'Pizza', 'NamedPizza', 'PizzaBase', and 'Topping'. 'CheesyPizza' is highlighted as a subclass of 'Pizza'. The main right-hand pane shows the 'Description: CheesyPizza' tab. In this pane, the 'Equivalent To' section contains the logical expression: $\text{Pizza} \text{ and } ((\text{hasBase some CheeseBorderBase}) \text{ or } (\text{hasTopping some Cheese}))$. This expression is circled in green. Below it, the 'SubClass Of' section lists 'Pizza' as a subclass. The 'Instances' section shows an instance named 'Pizza_39'.

Equivalent to

$\text{Pizza} \text{ and } ((\text{hasBase some CheeseBorderBase}) \text{ or } (\text{hasTopping some Cheese}))$

SubClass Of

Pizza

Instances

Pizza_39

Clasificación por rangos de valores

untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) : [C:\Users\RafaelValenciaGarcia\Dropbox\asignaturas\2018-2019\W...

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untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5)

Person > Child

Active ontology * Entities * Individuals by class * DL Query *

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: Child

owl:Thing

Location

Person

- Adult
- Baby
- Child
- Cousin
- Eldery
- GrandParent
- NephewNiece
- Parent
- PersonWithAtLeast2Children
- SocialCareBefeficiaryMurcia
- Spaniard
- Teenager
- TeenagerParent
- UncleAunt

Class Annotations Class Usage

Annotations: Child

Annotations +

Description: Child

Equivalent To +

Person and (age some xsd:int[>= 3 , <= 7])

SubClass Of +

Person

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

- Celia
- Hugo
- Isaac
- Teresita
- Victor

Reasoner active Show Inferences

Condiciones anidadas

untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) : [C:\Users\RafaelValenciaGarcia\Dropbox\asignaturas\2018-2019\W...

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< > untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) Search...

> Person > UncleAunt

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals
Classes Object properties Data properties

Class hierarchy: UncleAunt

- owl:Thing
 - Location
 - Person
 - Adult
 - Baby
 - Child
 - Cousin
 - Eldery
 - GrandParent
 - NephewNiece
 - Parent
 - TeenagerParent
 - PersonWithAtLeast2Children
 - SocialCareBeneficiaryMurcia
 - Spaniard
 - Teenager
 - TeenagerParent
 - UncleAunt**

Annotations: UncleAunt

Annotations +

Description: UncleAunt

Equivalent to +

- Person
 - and (hasSibling some (Person and (hasChild some Person)))

SubClass Of +

- Person

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

- Moises
- Rafa
- Teresa

Target for Key +

Reasoner active ☒ Show Inferences

Encadenamiento de properties

untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) : [C:\Users\RafaelValenciaGarcia\Dropbox\asignaturas\2018-2019\W...

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< > untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) Search...

hasCousin

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Object property hierarchy: hasCousin

owl:topObjectProperty

- hasAuntUncle
- hasChild
- hasCousin
- hasGrandChild
- hasGrandParent
- hasParent
- hasSibling
- isIn
- nacionality

Annotations Object Property Usage

Annotations: hasCousin

Annotations +

Characteristic

- ☐ Functional
- ☐ Inverse functional
- ☐ Transitive
- ☐ Symmetric
- ☐ Asymmetric
- ☐ Reflexive
- ☐ Irreflexive

Description: hasCousin

Equivalent To +

SubProperty Of +

Inverse Of +

Domains (intersection) +

Ranges (intersection) +

Disjoint With +

SuperProperty Of (Chain) +

hasParent o hasSibling o hasChild SubPropertyOf: hasCousin

Reasoner active Show Inferences

Functional property en data properties

The screenshot displays the Protégé ontology editor interface. The main window shows the 'identityCardNumber' data property being edited. The left sidebar contains a tree view of the ontology, with 'identityCardNumber' selected under 'Data properties'. The right sidebar shows the 'Description' tab for 'identityCardNumber', which is currently set to 'Functional'. Below this, there are sections for 'Equivalent To', 'SubProperty Of', 'Domains (intersection)', 'Ranges', and 'Disjoint With'. The 'Domains (intersection)' section shows a domain of 'Person', and the 'Ranges' section shows a range of 'xsd:string'. The bottom status bar indicates 'Reasoner active' and 'Show Inferences'.

untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) : [C:\Users\RafaelValenciaGarcia\Dropbox\asignaturas\2018-2019\W...

File Edit View Reasoner Tools Refactor Window Help

untitled-ontology-5 (http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5) Search...

identityCardNumber

Active ontology x Entities x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Data property hierarchy: identityCardNumber

owl:topDataProperty

age

identityCardNumber

identityCardNumber — http://www.semanticweb.org/alumno/ontologies/2015/10/untitled-ontology-5#identityCardNumber

Annotations Data Property Usage

Annotations: identityCardNumber

Annotations +

Characteristics: Functional

Description: identityCardNumber

Equivalent To +

SubProperty Of +

Domains (intersection) +

Person

Ranges +

xsd:string

Disjoint With +

Reasoner active Show Inferences

Bibliografía

- **Ontology Development 101: A Guide to Creating Your First Ontology**
 - http://protege.stanford.edu/publications/ontology_development/ontology101.pdf

Herramientas ontológicas 3

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BabelNet

Rodrigo Martínez Béjar

BabelNet is a multilingual lexicalized semantic network and ontology developed at the NLP group of the Sapienza University of Rome. BabelNet was automatically created by linking Wikipedia to the most popular computational lexicon of the English language, WordNet (Wikipedia)



BabelNet



English ▾ Translate into... ▾ 

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house English Translate into: Noun Verb Adj Adj

EN house noun



A dwelling that serves as living quarters for one or more families

dwelling • home

Concept architecture structure building



The members of a business organization that owns or operates one or more establishments

firm • business firm

Concept economics



The members of a religious community living together

Concept



The audience gathered together in a theatre or cinema

Concept

house English Translate into

WordNet 3.0

architecture structure building

See more

EXPLORE NETWORK

DEFINITIONS EXAMPLES RELATIONS SOURCES

English > More languages

IS A dwelling · building · home · housing · architectural structure

HAS PART loft · basement · door · drawing room · floor [+10 relations](#)

PART OF Brank · Ciboušov · Cowton railway station · Stadice · Partoltice [+8 relations](#)

HAS KIND sod house · beach house · boarding house · bungalow · cabin [+343 relations](#)

HAS INSTANCE preftiness · parainfluenza virus · smoke hole · Yung See San Fong House · Byron Howes House [+23K relations](#)

DERIVATION house

DESCRIBED BY SOURCE Brockhaus and Efron Encyclopedic Dictionary · Otto's encyclopedia · Encyclopædia Britannica 11th edition · Bible Encyclopedia of Archimandrite Nicephorus · Jewish Encyclopedia of Brockhaus and Efron

DIFFERENT FROM residence

● Semantically related form



CHANGE VIEW

TEXT

